The BCS-BEC Crossver in Ultracold Fermi Gases

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In this paper, I present a pedagogical overview of the Bardeen–Cooper–Schrieffer to Bose–Einstein condensation (BCS–BEC) crossover in ultracold Fermi gases, focusing on the role of tunable interactions via Feshbach resonances. Starting from the zero-range contact Hamiltonian and its mean-field solution, I show how to continuously connect overlapping Cooper pairs in the weak-coupling limit to tightly bound molecules in the strong-coupling limit, with the unitary regime marking maximal many-body interactions. I then review two pioneering experiments: the first observation of condensation on the BCS side and the imaging of quantized vortex lattices across the crossover in a rotating ^{6}Li gas, which provides unambiguous evidence of superfluidity.

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I. INTRODUCTION

The BCS–BEC crossover describes how a fermionic superfluid continuously evolves from the weak-coupling regime of overlapping Cooper pairs (BCS) to the strong-coupling regime of tightly bound bosonic molecules undergoing Bose–Einstein condensation (BEC). Shortly after Bardeen,

Cooper, and Schrieffer introduced their theory of superconductivity in 1957 [1] built on the idea of large, overlapping Cooper pairs, Schafroth, Butler, and Blatt [2] independently proposed an alternative picture in which tightly bound, non-overlapping bosonic pairs form and undergo BEC. These two approaches were originaly treated as unrelated: one describes weakly bound, many-body pairing, and the other describes preformed molecular bosons that condense in the usual BEC fashion.

Strong theoretical evidence and early experimental developments throughout the latter half of the twentieth century supported a unified picture in which these two limits connect smoothly. Eagles [3] first demonstrated at zero temperature that increasing the pairing strength drives a continuous evolution from BCS-like pairing to tightly bound dimers. Leggett [4] then showed that a self-consistent BCS wavefunction remains valid across the entire coupling range, and Nozières and Schmitt-Rink [5] extended this crossover above the critical temperature T_c, revealing that the condensation temperature itself smoothly evolves from the exponentially small BCS form to the ideal-Bose-gas BEC transition.

In the early twenty-first century, ultracold Fermi gases with interactions tunable via Feshbach resonances and advanced cooling and trapping techniques provided a simulation method of the BCS–BEC crossover, which is what this paper focuses on. I will began with a pedagogical introduction to many-body physics of Fermi gases and the methods for tuning interactions of these systems, followed by a breif theoretical overview of the BCS-BEC crossover. Finally, I will summarize two pioneering experiments: the observation of fermionic condensation across the crossover [6] and the imaging of vortex lattices as definitive evidence of superfluidity [7].

II. BACKGROUND

A. The Physics and Tunability of Ultracold Gases

The most commonly studied fermionic atoms in ultracold experiments are ^{40}K and ^{6}Li. To create a two-spin–component Fermi gas in ^{40}K, one applies a magnetic field that Zeeman-splits the $F = 9/2$ hyperfine manifold and then uses resonant radio-frequency pulses to transfer population between, for example, the $|F = 9/2, m_F = -9/2\rangle$ and $|F = 9/2, m_F = -7/2\rangle$ states, which serve as the $\uparrow$ and $\downarrow$ spin components [6].

To reach quantum degeneracy in a Fermi atomic gas and observe Fermi–Dirac statistics, the temperature must lie well below the Fermi temperature $T_F = \hbar^2 k_F^2/(2mk_B)$, where the Fermi

momentum k_F is related to the number density n by $k_F = (3\pi^2 n)^{1/3}$. Similarly, the thermal deBroglie wavelength $\lambda_{dB} = h/\sqrt{2\pi m k_B T}$ must become comparable to the mean spacing between particles k_F^{-1}. In typical ultracold–gas experiments, this condition corresponds to $T \lesssim 0.2\,T_F$, where T_F is on the order of 100nK [8].

At these ultralow temperatures, evaporative cooling is the method of choice, which is where sthe highest-energy atoms are continuously removed from a magnetic or optical dipole trap, and the remaining sample rethermalizes via elastic collisions. However, below roughly 1 μK only s-wave collisions remain energetically allowed—and by the Pauli exclusion principle s-wave scattering is forbidden between identical fermions. To cool fermions evaporatively, one therefore uses sympathetic cooling, which is where the fermions are co-trapped with a second species (or spin state) that undergoes s-wave collisions. As the "refrigerant" gas is cooled evaporatively, the fermions thermalize to the same temperature through elastic collisions. A common example is sympathetic cooling of ^{40}K atoms with a ^{87}Rb bosonic gas [9].

At ultralow temperatures in the dilute limit—when the interparticle spacing k_F^{-1} far exceeds the range of the interatomic potential r_0 so that $k_F r_0 \ll 1$—only s-wave scattering contributes, and the two-body interaction is completely characterized by the s-wave scattering length a. The low-energy scattering amplitude at momentum k is given by

$$f(k) = \frac{1}{k\cot\delta_0(k) - ik} \approx -\frac{1}{1/a + ik}\,,$$

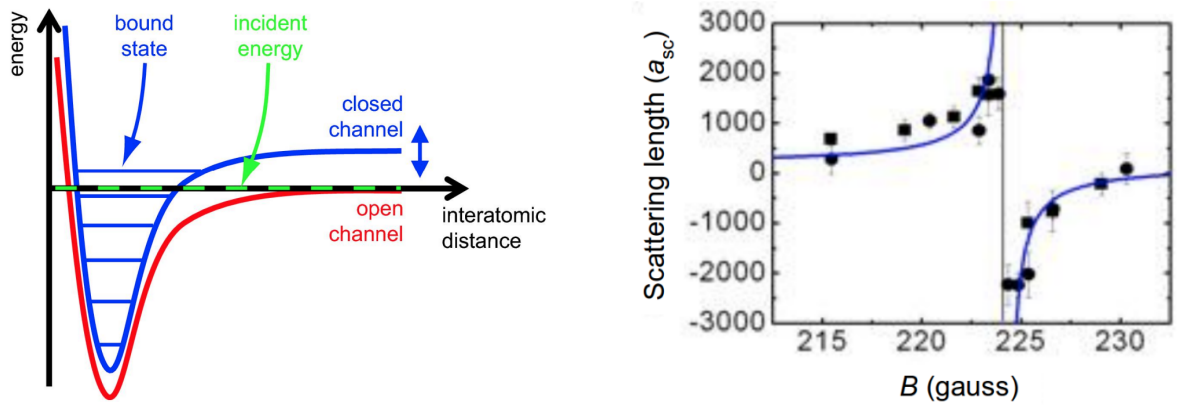
where, for $k \to 0$, $k\cot\delta_0 \approx -1/a$. The magnitude of a sets the interaction strength, while its sign indicates whether the interaction is attractive ($a < 0$) or repulsive ($a > 0$).

One can tune the scattering length a over a wide range using Feshbach resonance techniques[1] [10], which exploit a near-resonant coupling between an open-channel scattering state and a closed-channel bound state. In a two-spin-component Fermi gas, the open and closed channels may correspond, for example, to triplet and singlet spin configurations, respectively. At short interatomic separations, hyperfine and exchange interactions provide an off-diagonal coupling between these channels. By applying a homogeneous magnetic field, experimentalists shift their relative energies via the Zeeman effect; when the closed-channel bound state is brought into resonance with the open-channel continuum, the resulting hybridization causes the scattering length to diverge and change sign according to

$$a(B) = a_{\mathrm{BG}}\Big[1 - \frac{\Delta B}{B - B_0}\Big]\,. \tag{1}$$

[1] This paragraph on Feshbach resonance is adapted from a previous paper I submitted to Courseworks.

Here, $a_{\rm BG}$ is the background scattering length far from resonance, B_0 is the resonance field, and ΔB its width. Sweeping the magnetic field through B_0 therefore tunes the interaction from attractive ($a < 0$), through unitarity ($|a| \to \infty$), to repulsive ($a > 0$). Figure 1b shows experimental measurements of $a(B)$ for a two-spin-component ^{40}K gas.



(a) (b)

Figure 1: a) Schematic of the two-channel model for a Feshbach resonance. Atoms initially collide in the open channel potential $V_{\rm op}(r)$ (red curve). During the collision, the open channel is coupled to a closed channel potential $V_{\rm cl}(r)$ (blue curve) that supports a bound molecular state. [10] b) Feshbach control of interactions. Scattering length $a_{\rm sc}$ as a function of magnetic field B for collisions between ^{40}K atoms in the $|F = 9/2,\, m_F = -9/2\rangle$ and $|F = 9/2,\, m_F = -5/2\rangle$ states. The Feshbach resonance appears as a sign-changing divergence in a, enabling tuning from attractive ($a < 0$) through effectively non-interacting ($a \approx 0$) to repulsive ($a > 0$) interactions.[11]

In the dilute limit, the divergence of the two-body scattering length $|a| \to \infty$ defines the unitarity point, at which the s-wave phase shift at zero momentum reaches $\delta_0(0) = \pi/2$ and the scattering amplitude scales as $f(k) \sim 1/k$. By tuning a via a Feshbach resonance, one traverses the BCS–BEC crossover: the crossover is precisely located at unitarity, where $1/(k_F a) = 0$ (equivalently $|a| \to \infty$).

B. The BCS-BEC Crossover

To address the many-body problem, one replaces the true interatomic potential with a zero-range contact interaction $U = g\,\delta(\mathbf{r} - \mathbf{r}')$. In second quantization, the Hamiltonian then reads [8]

$$H = \sum_{\sigma=\uparrow,\downarrow} \int d^3r\, \psi_\sigma^\dagger \Big(-\frac{\nabla^2}{2m} - \mu\Big)\psi_\sigma \;-\; g(\Lambda) \int d^3r\, \psi_\uparrow^\dagger \psi_\downarrow^\dagger \psi_\downarrow \psi_\uparrow .$$

Here $\psi_\sigma(\mathbf{r})$ are the fermionic field operators, $-\frac{\nabla^2}{2m}-\mu$ is the single-particle grand-canonical operator with dispersion $\epsilon_k = \hbar^2 k^2/2m$, and the bare coupling $g(\Lambda)$ is chosen so that

$$\frac{m}{4\pi a} = -\frac{1}{g(\Lambda)} + \sum_{|\mathbf{k}|<\Lambda} \frac{1}{2\epsilon_k}$$

reproduces the physical scattering length a. Physically, Λ provides an ultraviolet cutoff—typically $\Lambda \sim 1/r_0 \gg k_F$—and this renormalization condition ensures that the contact interaction yields the correct low-energy two-body scattering amplitude, making all many-body observables depend only on the physical parameter a, not on Λ.

Fourier transforming the contact Hamiltonian into momentum space, applying a mean-field decoupling at $T = 0$, and performing the Bogoliubov transformation yields the variational BCS ground state [12]

$$|\mathrm{BCS}\rangle = \prod_{\mathbf{k}} \big(u_{\mathbf{k}} + v_{\mathbf{k}}\, c^\dagger_{\mathbf{k}\uparrow}\, c^\dagger_{-\mathbf{k}\downarrow}\big)|0\rangle, \tag{2}$$

with normalization $u_{\mathbf{k}}^2 + v_{\mathbf{k}}^2 = 1$, coherence factors $v_{\mathbf{k}}/u_{\mathbf{k}} = (\Delta/E_{\mathbf{k}} + \xi_{\mathbf{k}})$, single-particle energy $\xi_{\mathbf{k}} = \hbar^2 k^2/(2m) - \mu$, and quasiparticle dispersion $E_{\mathbf{k}} = \sqrt{\xi_{\mathbf{k}}^2 + \Delta^2}$. The order parameter Δ and chemical potential μ are fixed by the self-consistency ("gap" and "number") equations

$$\Delta = -\,g \sum_{\mathbf{k}} \frac{\Delta}{2E_{\mathbf{k}}}\,, \qquad n = 2\sum_{\mathbf{k}} \Big(1 - \frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}}\Big). \tag{3}$$

In the weak-coupling limit $1/(k_F a) \to -\infty$, these equations admit a solution with $\mu \approx E_F$ and $\Delta \ll E_F$, reproducing the BCS result $\Delta \sim \frac{8E_F}{e^2}\exp\!\big(-\frac{\pi}{2k_F|a|}\big)$. The corresponding real-space pair size follows from $\xi_{\mathrm{pair}} \sim \hbar v_F/(\pi\Delta)$, giving $\xi_{\mathrm{pair}} \sim k_F^{-1}\exp\!\big(\frac{\pi}{2k_F|a|}\big) \gg k_F^{-1}$. Thus, in the BCS regime Cooper pairs are large and overlapping. On the other hand, in the strong-coupling limit $1/(k_F a) \to +\infty$, one finds $\mu \to -E_b/2$ with $E_b = \hbar^2/(ma^2)$ and $\Delta \gg E_F$, so that the same state describes tightly bound dimers of size $\xi_{\mathrm{pair}} \sim a \ll k_F^{-1}$. At unitarity ($1/k_F a \to 0$), the pair size becomes on the order of the interparticle spacing, $\xi_{\mathrm{pair}} \sim k_F^{-1}$. Because both μ and Δ depend only on the single parameter $1/k_F a$ at $T = 0$, there is no singularity at unitarity, and the gap and number equations smoothly interpolate between the BCS and BEC limits. In summary, in the BCS limit ($1/(k_F a) \to -\infty$) there is no two-body bound state and pairing is purely a many-body effect; at unitarity ($1/(k_F a) = 0$) the system is maximally interacting; and in the BEC limit ($1/(k_F a) \to +\infty$) tightly bound fermion pairs behave as bosonic molecules.

At finite temperature the simple zero-temperature mean-field picture must be extended to involve fluctuations. Just above the critical temperature T_c, Nozières and Schmitt-Rink showed

that Gaussian fluctuations of the pairing field cause T_c to evolve smoothly from the exponentially small BCS result [13]

$$T_c \sim E_F \exp\left(-\frac{\pi}{2\,k_F|a|}\right) \quad (1/(k_F a) \to -\infty) \tag{4}$$

to the non-interacting BEC expression. In the BEC limit $1/(k_F a) \to +\infty$, the condensation temperature for bosonic dimers of mass $2m$ and density $n/2$ is $T_c = \left(n/2\,\zeta(3/2)\right)^{2/3} \pi/m \simeq 0.22\,T_F$. Above T_c, one defines a pairing temperature T^* at which two-body correlations become significant; on the BCS side $T^* \approx T_c$, while on the BEC side $T^* \sim \frac{E_b}{\ln(E_b/E_F)}$ [8]. Between T_c and T^* the system can exhibit a "pseudogap," in which fermions form incoherent pairs without long-range superfluid order. Throughout the entire crossover, the single dimensionless coupling $1/(k_F a)$ continues to govern both the superfluid transition and the character of the normal state up to temperatures of order T_F. Figure 2 displays the phase diagram of the BSC-BEC crossover.

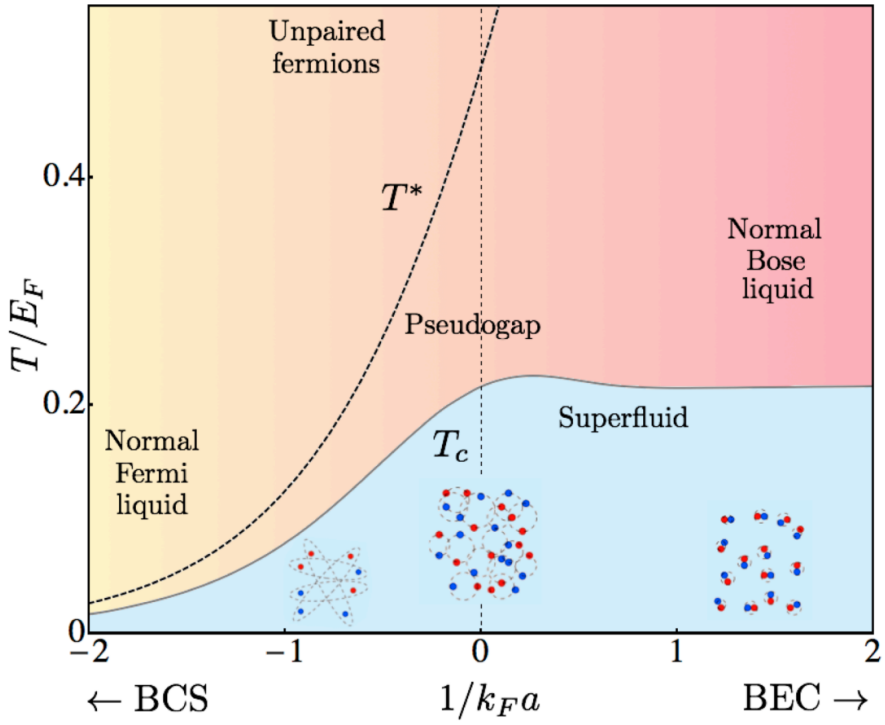


Figure 2: Phase diagram for BCS-BEC crossover as a function of T/T_F and $1/k_F a$ [8]

III. EXPERIMENTS

A. Observation of Resonance Condensation of Fermionic Atom Pairs

The first experiment I will summarize was performed in 2004 at the University of Colorado, Boulder, where the first observation of condensation on the BCS side ($a < 0$) was achieved [6]. Previously, fermionic condensation had only been seen in the BEC molecular limit because, on the BEC side, tightly bound molecules expand ballistically and reveal their momentum distribution in time-of-flight images, whereas on the BCS side Cooper pairs are too weakly bound and many-body interactions disrupt the expansion. To overcome this, the authors introduced a pair-projection technique: the magnetic field is rapidly swept from the BCS side across the Feshbach resonance to the BEC side, slow enough to convert pairs into molecules but fast enough to prevent significant motion or collisions in the trap. In their setup, an equal mixture of $|F = 9/2, m_F = -9/2\rangle$ and $|F = 9/2, m_F = -7/2\rangle$ ^{40}K atoms was evaporatively cooled to quantum degeneracy and confined in an optical dipole trap. The gas was then prepared on the BCS side before applying the projection sweep. Figure 3 shows the resulting time-of-flight images at increasing detunings ΔB (left to right). The pronounced central peak in the leftmost image reveals a macroscopic occupation of the zero-momentum state by fermion pairs; as ΔB increases at fixed temperature T, the ratio T/T_c rises and the condensate fraction, and thus the height of the central peak, diminishes.

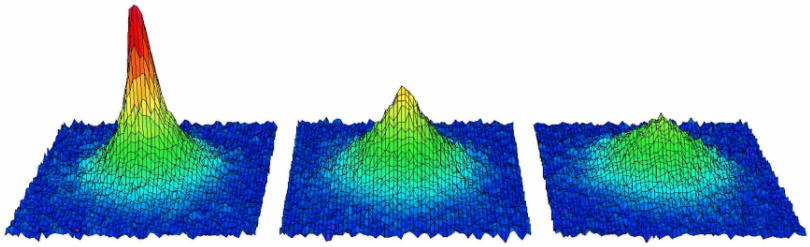


Figure 3: Time-of-flight expansion of fermionic pairs for $\Delta B = 0.12G, 0.25G, 0.55G$ from left to right at $T/T_F = 0.07$. The central peaks indicate condensation. [6]

B. Observation of Superfluidity

In the next experiment conducted at MIT in 2005 [7], the researchers extended studies of the BCS–BEC crossover by imaging vortex lattices on both sides of resonance in a strongly interacting, rotating ultracold Fermi gas. Quantized vortices reveal the superfluid's macroscopic wavefunction

because superflow is irrotational, angular momentum is carried only by discrete vortex lines. Here, fermionic ^{6}Li atoms were first sympathetically cooled by ^{23}Na in a magnetic trap and then transferred into a high-field optical dipole trap. An equal mixture of the two lowest hyperfine states of ^{6}Li was prepared near the Feshbach resonance on the BEC side. Evaporative cooling below T_c produced a molecular BEC, which was then set into rotation by a two-beam repulsive stirrer. After equilibration, the trap was turned off and the cloud expanded for time-of-flight imaging, directly revealing the vortex lattice in momentum space. To observe vortices on the BCS side, where many-body effects would otherwise disrupt vortex signatures during expansion, the magnetic field was swept across resonance during time of flight to convert Cooper pairs into stable molecules with the same center-of-mass momentum. Figure 4 displays these time-of-flight images on the BEC side, at unitarity, and on the BCS side, each showing clear vortex lattices and thus unambiguous superfluidity throughout the crossover.

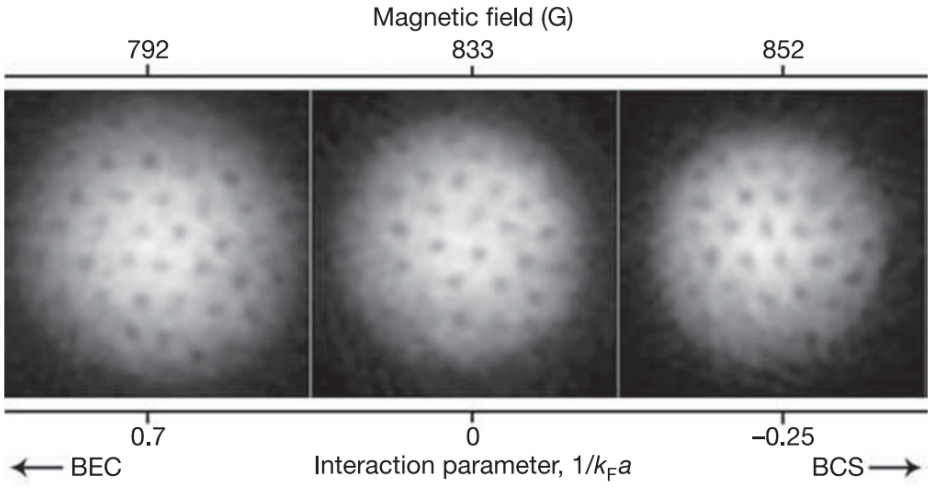


Figure 4: Vortex lattices throughout the BCS-BEC crossover. The field was ramped to the BEC side during expansion [7].

IV. CONCLUSION

In this paper I have reviewed the theoretical foundations and experimental realizations of the BCS–BEC crossover in ultracold Fermi gases. Starting from the zero-range contact Hamiltonian and its mean-field (BCS) solution, I showed how the self-consistent gap and number equations yield a smooth evolution from large, overlapping Cooper pairs in the weak-coupling limit to tightly bound molecular dimers in the strong-coupling limit, with the unitary point $|a| \to \infty$ marking the

strongest interactions. I then summarized two landmark experiments: the projection technique that revealed condensation on the BCS side in a ^{40}K gas, and the imaging of quantized vortex lattices across the crossover in a rotating ^{6}Li gas, providing strong evidence of superfluidity throughout the entire regime.

Looking forward, the tunability of ultracold atoms opens many exciting directions. Mixtures of fermions, such as ^{6}Li–and ^{40}K, with unequal masses, or three hyperfine components may realize analogues of color superconductivity or exotic phase-separated and coexistence states predicted in dense quark matter. Higher-angular-momentum (p-wave) Feshbach resonances promise to illuminate anisotropic superfluid phases and topological transitions. Finally, loading interacting fermions into optical lattices brings us close to realizing high-T_c cuprate phase diagrams, including pseudogap behavior and d-wave pairing. In all these cases, ultracold Fermi gases offer a highly controllable platform to simulate and deepen our understanding of strongly correlated fermionic matter across physics from condensed matter to nuclear and astrophysics [14].

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